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"Image resolution"

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Problem Definition:

Many of the images are old and some were not captured properly, and many of them have issues like noise or unsharp or low-quality It causes the image to lose its meaning or clarity, which causes people find a significant challenge to Correct and fix images.
As a consequence, many people have been trying to find ways to improve images quality. in order to restore an image, these flaws must be fixed in order to return it to its original, or at least visually appealing, state. As a result, our project aims to improve image resolution by using image processing methods.

Project goal:

The primary goal of this project is to fix images using restoration techniques, to rejuvenate aging and damaged photographs , make them clearer.
We aim is to processing these images, enhancing clarity, removing noise, addressing imperfections and improving image quality.

Processing Identification:

1-Image Filtering: Utilize filtering techniques, such as Median filtering, to reduce or remove noise and enhance the overall quality of the images.

2- Assessment of Image Quality: This involves evaluating the overall quality of the old image, including factors such as fading, loss of detail, discoloration, and blurriness.

3-Selection of Restoration Techniques: Based on the identified imperfections and restoration goals, appropriate techniques are chosen. These may include sharpening, noise reduction, contrast enhancement, scratch removal, color correction, and unsharp image restoration.

METHODOLOGY using Median Filter

Image with Salt and pepper noise is an impulse type of noise in images which is generally caused by errors in data transmission, failure in memory cell or analog-to-digital converter errors, which can significantly deteriorate the quality of an image.

The median filter is an effective method that can fix the Salt and pepper noise in image. Specifically, the median filter replaces a pixel by the median, instead of the average, of all pixels in a neighborhood Average (or mean) filtering is a method of 'smoothing' images by reducing the amount of intensity variation between neighbouring pixels.





Code using Median Filter

```
clc;
clear;
close all;

% Load image
I = imread('BO.jpg');
if numel(size(I)) > 2
    Im = rgb2gray(I);
else
    Im = I; % Image is already grayscale
end

noisy = imnoise(Im, 'salt & pepper', 0.1);

[m, n] = size(noisy);

output = zeros(m, n);
output = uint8(output);

for i = 1:m
    for j = 1:n
        xmin = max(1, i - 1);
        xmax = min(m, i + 1);
        ymin = max(1, j - 1);
        ymax = min(n, j + 1);
        temp = noisy(xmin:xmax, ymin:ymax);
        output(i, j) = median(temp(:));
    end
end

figure(1);
set(gcf, 'Position', get(0, 'ScreenSize'));
subplot(131), imshow(I), title('Original Image');
subplot(132), imshow(output), title('Output of Median Filter');
```

Code results using Median Filter

Original Image



Output of Median Filter



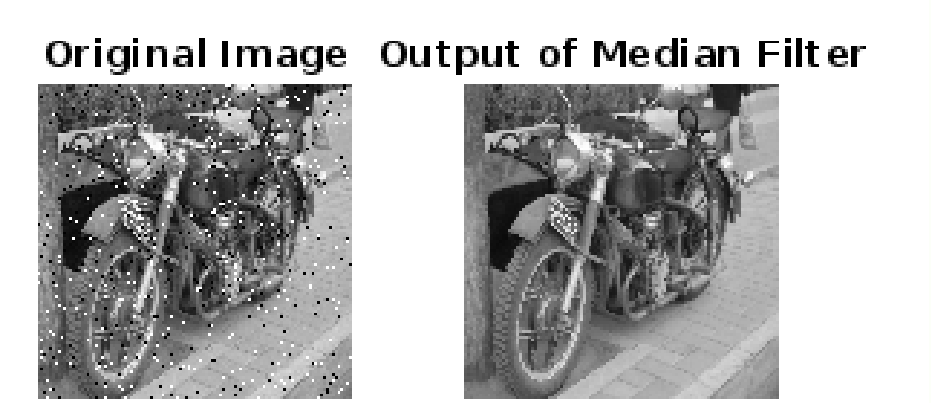
Original Image



Output of Median Filter



COMPARISION RESULTS



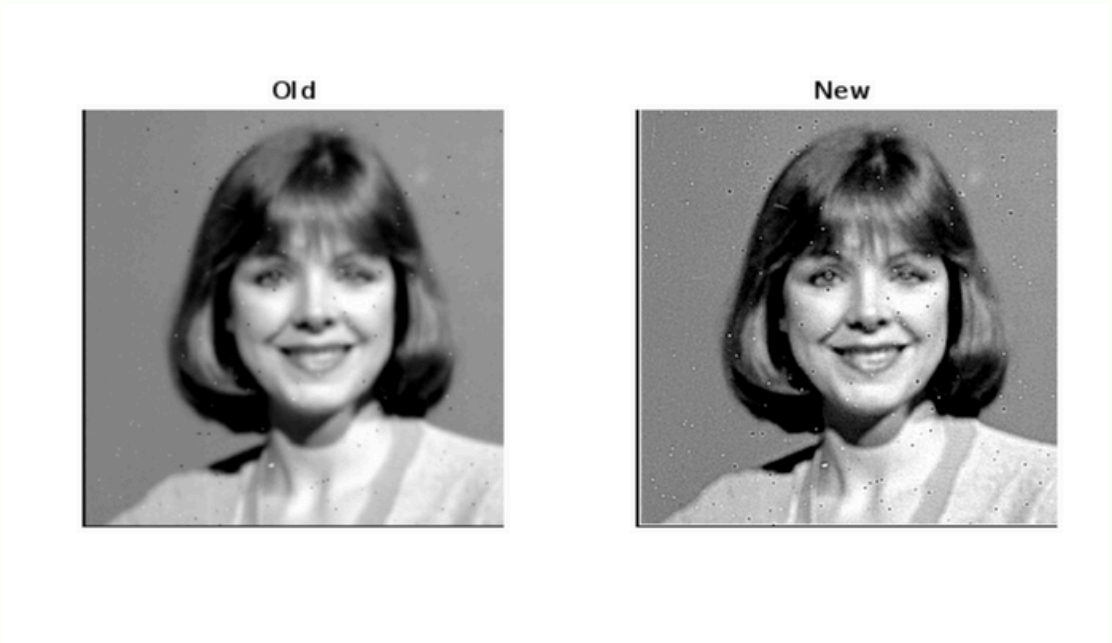
Noise Level: The amount of salt-and-pepper noise present is reduced in the processed image.
Clarity: The processed image exhibits improved clarity and detail compared to the noisy image.

Noise Density (0.01):
which is the proportion of pixels that will be affected by salt and pepper noise.
Specifically, a density of 0.01 means that approximately 1% of the pixels in the image will be affected by salt and pepper noise.

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METHODOLOGY USING UNSHARP IMAGE

The problem of restoring an unsharp image refers to improving the quality of an image that lacks sharpness in its details . When an image is unsharp, it may appear blurry and lack the fine details that make it clear and sharp.



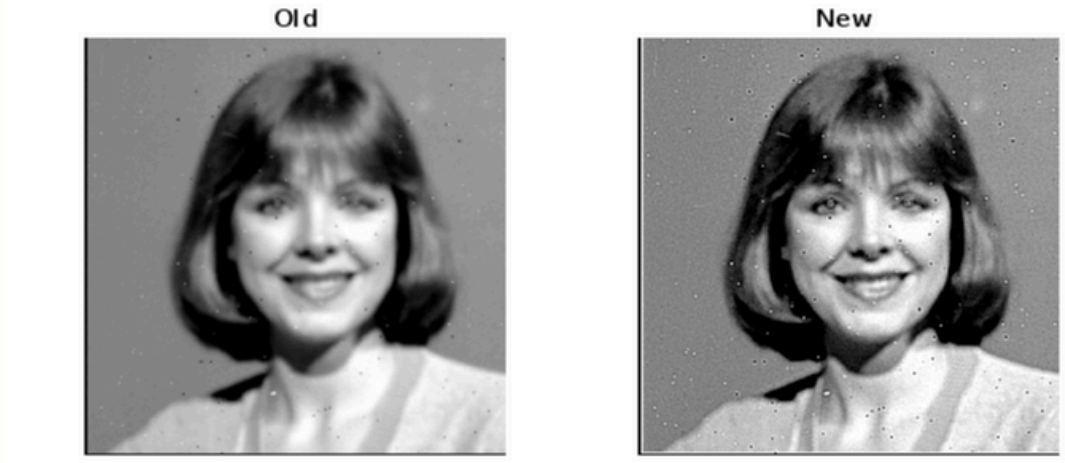
Code using unsharp

/MATLAB Drive/project/lama.m

```
1   % Read the image
2   image = imread('/MATLAB Drive/لقطة الشاشة ١٤٤٥-١١-٠١ في ٩.٥٠.٣١ ص.png');
3
4   % Display the original image
5   subplot(1, 2, 1);
6   imshow(image);
7   title('Old');
8
9   % Aplay
10  sharpened_Image = imsharpen(image,Radius=1.5,Amount=6);
11
12  % Display the processed image
13  subplot(1, 2, 2);
14  imshow(sharpened_Image);
15  title('New ');
16
```

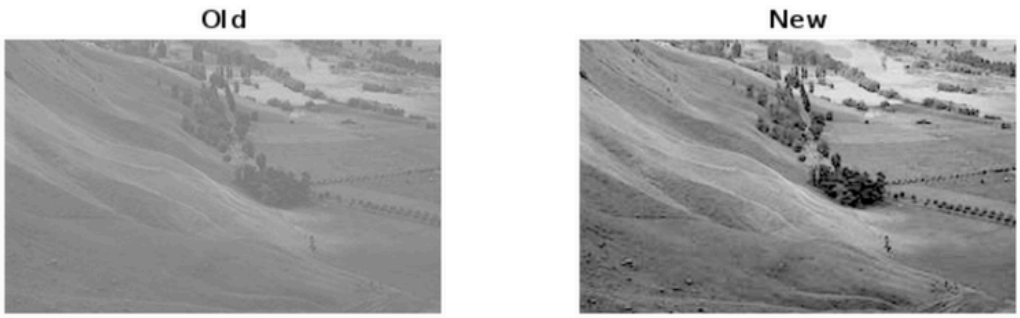
COMPARISION RESULTS

When comparing old and new images using the Unsharp feature, the differences are strikingly evident. the tool excels in enhancing the fine details that might have been lost or blurred in the old image, rejuvenating its clarity and sharpness.



METHODOLOGY IMAGE CONTRAST

The contrast method involves adjusting intensity values to enhance an image's visual appeal and restore its contrast. This technique widely used in image processing, improves visibility of details, ultimately elevating the overall quality and impact of the image.



Code using contrast

lama.m × Lama2.m × Figure 1 ×

/MATLAB Drive/Lama2.m

```
% Read the image
original_image = imread('/MATLAB Drive/لقطة الشاشة ١٤٤٥-١١-٠١ في ١٢.٠١.٥١ م.png');

% Display the original
subplot(1, 2, 1);
imshow(original_image);
title('Old');

% Convert the original image to double format
original_image_double = im2double(original_image);

% Define lower and upper percentiles for contrast
lowerPercentile = 0.9;
highPercentile = 100;

% Compute the intensity limits for contrast stretching
limits = stretchlim(original_image, [lowerPercentile/100, highPercentile/100]);

% Apply contrast stretching to restore image contrast
restoredImage = imadjust(original_image, limits, []);

% Define gamma value for gamma correction  Apply gamma correction
gamma = 0.4;
enhancedImage = imadjust(restoredImage, [], [], gamma);

% Display the new image
subplot(1, 2, 2);
imshow(enhancedImage);
title('New');

```

COMPARISION RESULTS

When comparing old and new images using the Contrast feature, the disparities are immediately apparent. The tool effectively enhances the visual appeal of the old image by amplifying the contrast, thereby revitalizing its clarity and sharpness.

Old



New



METHODOLOGY IMAGE CONTRAST & Sharping

The methodology for image contrast and sharpening involves two key techniques. Image contrast aims to improve visual quality by adjusting pixel intensities through methods such as histogram equalization. This technique enhance contrast and reveal more details. On the other hand, image sharpening techniques enhance edge and fine details, improving perceived sharpness.



the code using image Contrast and Sharping

```
folder = '/MATLAB Drive/project/images';

fileList = dir(fullfile(folder, '*.jpg'));

maxNumImages = 2;

for i = 1:min(numel(fileList), maxNumImages)
    filename = fullfile(folder, fileList(i).name);
    image = imread(filename);

    figure('Position', [200, 200, 800, 600]);

    subplot(1, 3, 1);
    imshow(image);
    title('Original image', 'FontSize', 6);

    image1 = im2double(image);

    radius = 5;
    amount = 1.5;
    enhanced1 = imsharpen(image1, 'Radius', radius, 'Amount', amount);

    subplot(1, 3, 2);
    imshow(enhanced1);
    title('Image Sharpening', 'FontSize', 6);

    image2 = im2double(image);

    low_in = 0.1;
    high_in = 0.9;
    enhanced2 = imadjust(image2, stretchlim(image2, [low_in, high_in]), []);

    subplot(1, 3, 3);
    imshow(enhanced2);
    title('Contrast Stretching', 'FontSize', 6);
end
```

COMPARISION RESULTS

The comparison between image contrast enhancement and sharpening techniques reveals their distinct contributions to image quality. Contrast enhancement methods like contrast stretching improve overall contrast, while sharpening techniques enhance edge and detail visibility. Both methods significantly enhance visual appeal and detail perception in images.





Conclusion

To conclude, image restoration plays a vital role in the project's objective of preserving and revitalizing deteriorated or damaged old photographs. By utilizing advanced restoration techniques like scratch removal or noise, and clarity enhancement, the images are restored to their original or improved condition.
This process not only enhances their visual attractiveness but also safeguards their cultural, historical, and personal significance for future generations to appreciate.

References

1 Salt and Pepper Noise Dataset: noisy image. (n.d.). In https://www.kaggle.com/datasets/rajneesh231/salt-and-pepper-noise-images/data.

2 https://homepages.inf.ed.ac.uk/rbf/HIPR2/images/wom1noi1.gif

3 https://www.researchgate.net/figure/Image-with-bad-contrast_fig5_341251073

4 https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=442c6559f5319db78d9888d23277ea540a3e0bb8

5 https://www.nv5geospatialsoftware.com/docs/SharpenImages.html